

The Three-Dimensional Spherical Quantum Harmonic Oscillator

Derivation Companion to “A Theoretical Exploration”

Aurora M. Bowman

Advisor: Dr. Souvik Das
Florida Institute of Technology

April 24, 2026

1 The Schrödinger Equation in Spherical Coordinates

We begin with the 3D time-independent Schrödinger equation in Cartesian coordinates:

$$-\frac{\hbar^2}{2m}\nabla^2\Psi + \frac{1}{2}m\omega^2(x^2 + y^2 + z^2)\Psi = E\Psi \quad (1)$$

where m is the particle's mass, ω is the angular frequency, E is the total energy, and the potential $V(x, y, z) = \frac{1}{2}m\omega^2(x^2 + y^2 + z^2)$. Note that $x^2 + y^2 + z^2 = r^2$. Since the potential depends only on r (now $V(r)$), it is natural to work in spherical coordinates. The Laplacian ∇^2 in spherical coordinates is

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}. \quad (2)$$

We assume a separable solution:

$$\Psi(r, \theta, \phi) = R(r)\Theta(\theta)\Phi(\phi). \quad (3)$$

Substituting Equation (2) into Equation (1) and dividing both sides by $\Psi = R\Theta\Phi$, we obtain

$$-\frac{\hbar^2}{2m}\left[\frac{1}{Rr^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{1}{r^2\Theta\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \frac{1}{r^2\Phi\sin^2\theta}\frac{d^2\Phi}{d\phi^2}\right] + \frac{1}{2}m\omega^2r^2 = E. \quad (4)$$

Multiplying through by $\frac{2m}{\hbar^2}$ gives

$$-\left[\frac{1}{Rr^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{1}{r^2\Theta\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \frac{1}{r^2\Phi\sin^2\theta}\frac{d^2\Phi}{d\phi^2}\right] + \frac{m^2\omega^2}{\hbar^2}r^2 = \frac{2mE}{\hbar^2}. \quad (5)$$

Rearrange:

$$\frac{1}{R}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{2mE}{\hbar^2}r^2 - \frac{m^2\omega^2}{\hbar^2}r^4 = -\left[\frac{1}{\Theta\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \frac{1}{\Phi\sin^2\theta}\frac{d^2\Phi}{d\phi^2}\right]. \quad (6)$$

The left-hand side depends only on r , while the right-hand side depends only on θ, ϕ . Both sides must equal a constant. Due to the requirement that $\Theta(\theta)$ and $\Phi(\phi)$ must be finite everywhere on the sphere, we denote the constant as $\ell(\ell + 1)$.

This gives the angular equation:

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta}{d\theta}\right) + \frac{1}{\sin^2\theta}\frac{d^2\Phi}{d\phi^2} + \ell(\ell + 1)\Theta\Phi = 0. \quad (7)$$

The radial equation becomes

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{dR}{dr}\right) + \frac{2mE}{\hbar^2}R - \frac{m^2\omega^2}{\hbar^2}r^2R - \frac{\ell(\ell + 1)}{r^2}R = 0. \quad (8)$$

2 Angular Solutions of the 3D Quantum Harmonic Oscillator

From separation of variables, the angular part of the Schrödinger equation is

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \frac{1}{\sin^2 \theta} \frac{d^2 \Phi}{d\phi^2} + \ell(\ell+1)\Theta\Phi = 0. \quad (9)$$

Step 1: Separate θ and ϕ parts

Divide by $\Theta\Phi$:

$$\frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \ell(\ell+1) = -\frac{1}{\Phi \sin^2 \theta} \frac{d^2 \Phi}{d\phi^2}. \quad (10)$$

Since each side depends on different variables, both must equal a constant. Let this constant be m^2 . Then we obtain two separate equations:

$$\frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \ell(\ell+1) = m^2, \quad (11)$$

$$-\frac{1}{\Phi \sin^2 \theta} \frac{d^2 \Phi}{d\phi^2} = m^2. \quad (12)$$

Step 2: Solve for $\Phi(\phi)$

Multiplying both sides of Equation (12) by Φ gives

$$-\frac{1}{\sin^2 \theta} \frac{d^2 \Phi}{d\phi^2} = m^2 \Phi. \quad (13)$$

Since this equation must depend only on ϕ , the θ -dependent factor must cancel, yielding

$$\frac{d^2 \Phi}{d\phi^2} + m^2 \Phi = 0. \quad (14)$$

This equation is a standard linear homogeneous second-order differential equation with a known solution:

$$\Phi(\phi) = Ae^{im\phi} + Be^{-im\phi}. \quad (15)$$

Since $e^{im\phi}$ and $e^{-im\phi}$ are linearly independent solutions, any linear combination can be expressed by re-defining constants. Thus, we may choose one of the independent solutions and absorb constants into a single coefficient:

$$\Phi(\phi) = Ce^{im\phi}, \quad (16)$$

where C is a new constant.

This choice is equivalent to setting either $A = 0$ or $B = 0$, and does not affect the generality of the solution since each term individually satisfies the differential equation and boundary conditions.

We now impose the normalization condition to determine the constant C :

$$\int_0^{2\pi} |\Phi(\phi)|^2 d\phi = 1. \quad (17)$$

Substituting $\Phi(\phi) = Ce^{im\phi}$ gives

$$|\Phi(\phi)|^2 = |C|^2, \quad (18)$$

since $|e^{im\phi}|^2 = 1$. So,

$$|C|^2 \int_0^{2\pi} d\phi = |C|^2 (2\pi) = 1, \quad (19)$$

which yields

$$C = \frac{1}{\sqrt{2\pi}}. \quad (20)$$

Normalized solution:

$$\boxed{\Phi(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi}} \quad (21)$$

Single-valuedness of the wavefunction requires

$$\Phi(\phi + 2\pi) = \Phi(\phi).$$

After canceling the constant on both sides, this gives

$$e^{im(\phi+2\pi)} = e^{im\phi} \Rightarrow e^{2\pi im} = 1 \quad (22)$$

Therefore, m must be an integer:

$$m = 0, \pm 1, \pm 2, \dots \quad (23)$$

We write this more simply as

$$\boxed{m \in \mathbb{Z}} \quad (24)$$

m is the **magnetic quantum number** and defines the orientation of the wavefunction in space.

Step 3: Solve for $\Theta(\theta)$

Substituting the separation constant m^2 into the angular equation gives

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + \left[\ell(\ell + 1) - \frac{m^2}{\sin^2 \theta} \right] \Theta = 0. \quad (25)$$

To simplify this equation, we introduce the substitution

$$x = \cos \theta. \quad (26)$$

Using the chain rule,

$$\frac{d}{d\theta} = -\sin \theta \frac{d}{dx},$$

the differential equation transforms into

$$(1 - x^2) \frac{d^2 \Theta}{dx^2} - 2x \frac{d\Theta}{dx} + \left[\ell(\ell + 1) - \frac{m^2}{1 - x^2} \right] \Theta = 0. \quad (27)$$

This is the associated Legendre differential equation. Its solutions are the **associated Legendre functions**:

$$\boxed{\Theta(\theta) \propto P_\ell^m(\cos \theta)} \quad (28)$$

Note: The normalization constant for $\Theta(\theta)$ is not determined in this derivation, but it can be found.

Remark: Legendre and Associated Legendre Polynomials

The functions $P_\ell(x)$ are called Legendre polynomials, which arise as solutions to the differential equation

$$(1 - x^2)y'' - 2xy' + \ell(\ell + 1)y = 0. \quad (29)$$

They form a set of orthogonal polynomials on the interval $[-1, 1]$ and frequently appear in problems with spherical symmetry.

The associated Legendre functions $P_\ell^m(x)$ generalize these solutions to include the parameter m , and are defined (for $m \geq 0$) by

$$P_\ell^m(x) = (1 - x^2)^{m/2} \frac{d^m}{dx^m} P_\ell(x). \quad (30)$$

In this problem, they describe how the wavefunction varies with the polar angle θ and ensure that the angular solution remains finite at the poles.

To ensure that the solution remains finite at $\theta = 0$ and $\theta = \pi$, we require

$$\boxed{\ell = 0, 1, 2, \dots \quad |m| \leq \ell} \quad (31)$$

ℓ can be defined as the **angular momentum quantum number** and determines the angular momentum and shape of the wavefunction.

3 Radial Solution of the 3D Quantum Harmonic Oscillator

Step 1: Identify the radial equation

After separating angular and radial parts of the 3D spherical Schrödinger equation in Section (1), the radial equation is identified as

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{2mE}{\hbar^2} R - \frac{m^2 \omega^2}{\hbar^2} r^2 R - \frac{\ell(\ell+1)}{r^2} R = 0. \quad (32)$$

.

Step 2: Expand the radial derivative

Expand the derivative using the product rule:

$$\frac{d}{dr} (r^2 R') = r^2 R'' + 2r R'. \quad (33)$$

Substitute into the equation:

$$\frac{1}{r^2} (r^2 R'' + 2r R') + \left(\frac{2mE}{\hbar^2} - \frac{m^2 \omega^2}{\hbar^2} r^2 - \frac{\ell(\ell+1)}{r^2} \right) R = 0. \quad (34)$$

Expand by multiplying the first two terms by $\frac{1}{r^2}$:

$$R'' + \frac{2}{r} R' + \left(\frac{2mE}{\hbar^2} - \frac{m^2 \omega^2}{\hbar^2} r^2 - \frac{\ell(\ell+1)}{r^2} \right) R = 0. \quad (35)$$

Step 3: Eliminate the first derivative

The $\frac{2}{r} R'$ term makes the equation harder to solve. We eliminate it by defining:

$$u(r) = r R(r). \quad (36)$$

Then

$$R = \frac{u}{r}. \quad (37)$$

Now compute derivatives:

First derivative

$$R' = \frac{d}{dr} \left(\frac{u}{r} \right) \quad (38)$$

$$= \frac{u'r - u}{r^2} \quad (39)$$

$$= \frac{u'}{r} - \frac{u}{r^2}. \quad (40)$$

Second derivative

$$R'' = \frac{d}{dr} \left(\frac{u'}{r} - \frac{u}{r^2} \right) \quad (41)$$

$$= \frac{u''}{r} - \frac{u'}{r^2} - \left(\frac{u'}{r^2} - \frac{2u}{r^3} \right) \quad (42)$$

$$= \frac{u''}{r} - \frac{2u'}{r^2} + \frac{2u}{r^3}. \quad (43)$$

Now we compute

$$R'' + \frac{2}{r}R' = \left(\frac{u''}{r} - \frac{2u'}{r^2} + \frac{2u}{r^3} \right) + \frac{2}{r} \left(\frac{u'}{r} - \frac{u}{r^2} \right) \quad (44)$$

$$= \frac{u''}{r} - \frac{2u'}{r^2} + \frac{2u}{r^3} + \frac{2u'}{r^2} - \frac{2u}{r^3}. \quad (45)$$

Everything cancels except

$$R'' + \frac{2}{r}R' = \frac{u''}{r}. \quad (46)$$

Substitute back:

$$\frac{u''}{r} + \left(\frac{2mE}{\hbar^2} - \frac{m^2\omega^2}{\hbar^2}r^2 - \frac{\ell(\ell+1)}{r^2} \right) \frac{u}{r} = 0. \quad (47)$$

Multiply by r for simplification:

$$u'' + \left(\frac{2mE}{\hbar^2} - \frac{m^2\omega^2}{\hbar^2}r^2 - \frac{\ell(\ell+1)}{r^2} \right) u = 0. \quad (48)$$

Key result: we now have a simpler second-order ODE with no first derivative.

Step 4: Introduce a dimensionless variable

Define

$$\rho = \frac{r}{a}. \quad (49)$$

a is considered the "oscillator length" or "characteristic length scale". We choose a so that the r^2 term simplifies nicely because $\rho \approx 1$ corresponds to $r \approx a$. This arises from balancing potential and kinetic energy:

$$\frac{\hbar^2}{2ma^2} \approx \frac{1}{2}m\omega^2a^2 \quad (50)$$

$$a \approx \sqrt{\frac{\hbar}{m\omega}} \quad (51)$$

This choice ensures:

$$\frac{m^2\omega^2}{\hbar^2}a^2 = 1. \quad (52)$$

Now transform derivatives:

$$\frac{d}{dr} = \frac{d\rho}{dr} \frac{d}{d\rho} = \frac{1}{a} \frac{d}{d\rho}, \quad (53)$$

$$\frac{d^2}{dr^2} = \frac{1}{a^2} \frac{d^2}{d\rho^2}. \quad (54)$$

Substitute $r = a\rho$ into the equation:

$$\frac{d^2 u}{d\rho^2} + \left(\frac{2mE}{\hbar^2} a^2 - \rho^2 - \frac{\ell(\ell+1)}{\rho^2} \right) u = 0. \quad (55)$$

Since

$$\frac{2mE}{\hbar^2} a^2 = \frac{2E}{\hbar\omega}, \quad (56)$$

define

$$\epsilon = \frac{E}{\hbar\omega}. \quad (57)$$

Final form:

$$u'' + \left(2\epsilon - \rho^2 - \frac{\ell(\ell+1)}{\rho^2} \right) u = 0. \quad (58)$$

Step 5: Analyze asymptotic behavior

Large ρ :

$$u'' - \rho^2 u \approx 0 \Rightarrow u \sim e^{-\rho^2/2}. \quad (59)$$

Small ρ :

$$u'' - \frac{\ell(\ell+1)}{\rho^2} u \approx 0 \Rightarrow u \sim \rho^{\ell+1}. \quad (60)$$

Both of these limits motivate us to use

$$u(\rho) = \rho^{\ell+1} e^{-\rho^2/2} v(\rho). \quad (61)$$

Step 6: Compute derivatives

Let

$$u = c(\rho)v(\rho), \quad c = \rho^{\ell+1} e^{-\rho^2/2}. \quad (62)$$

Then,

$$u' = c'v + cv', \quad (63)$$

$$u'' = c''v + 2c'v' + cv''. \quad (64)$$

Compute derivatives of $c(\rho)$:

$$c(\rho) = \rho^{\ell+1} e^{-\rho^2/2}. \quad (65)$$

Computing the first derivative:

$$c' = (\ell+1)\rho^\ell e^{-\rho^2/2} - \rho^{\ell+2} e^{-\rho^2/2} \quad (66)$$

$$= c \left(\frac{\ell+1}{\rho} - \rho \right), \quad (67)$$

Computing the second derivative:

Using

$$\frac{c''}{c} = \left(\frac{c'}{c}\right)^2 + \frac{d}{d\rho} \left(\frac{c'}{c}\right), \quad (68)$$

we compute:

$$\left(\frac{c'}{c}\right)^2 = \frac{(\ell+1)^2}{\rho^2} - 2(\ell+1) + \rho^2, \quad (69)$$

$$\frac{d}{d\rho} \left(\frac{c'}{c}\right) = -\frac{\ell+1}{\rho^2} - 1. \quad (70)$$

Divide c' and c'' by c for simplification:

$$\frac{c'}{c} = \frac{\ell+1}{\rho} - \rho, \quad (71)$$

$$\frac{c''}{c} = \frac{\ell(\ell+1)}{\rho^2} - (2\ell+3) + \rho^2. \quad (72)$$

Substitute into the equation and simplify carefully. After cancellation of terms, we obtain:

$$v'' + \left(\frac{2(\ell+1)}{\rho} - 2\rho\right) v' + (2\epsilon - 2\ell - 3) v = 0. \quad (73)$$

Step 7: Convert to an equation with known solution

Let

$$x = \rho^2. \quad (74)$$

Using the chain rule:

$$\frac{d}{d\rho} = 2\rho \frac{d}{dx}, \quad (75)$$

$$\frac{d^2}{d\rho^2} = 2 \frac{d}{dx} + 4\rho^2 \frac{d^2}{dx^2}. \quad (76)$$

Substitute into the equation and simplify to obtain:

$$xv'' + \left(\ell + \frac{3}{2} - x\right) v' + kv = 0, \quad (77)$$

where

$$k = \epsilon - \frac{2\ell+3}{2}. \quad (78)$$

This is the associated Laguerre equation.

Remark: Associated Laguerre Equation

The differential equation

$$xv'' + (\alpha + 1 - x) v' + kv = 0 \quad (79)$$

is known as the associated Laguerre equation.

Its solutions are the associated Laguerre polynomials

$$v(x) = L_k^\alpha(x), \quad (80)$$

which are a family of special functions that frequently arise when solving problems with radial symmetry.

Key properties of associated Laguerre polynomials:

- $k = 0, 1, 2, \dots$ must be a non-negative integer for the solution to be a polynomial.
- $\alpha > -1$ is a parameter determined by the problem (here, $\alpha = \ell + \frac{1}{2}$).
- These polynomials are finite and therefore terminate to physically acceptable, normalizable wavefunctions.

In this problem, identifying the differential equation as a Laguerre equation allows us to immediately write down the solution and determine the allowed energy values from the requirement that k be a non-negative integer.

Step 8: Identify the radial solution

$$v(x) = L_k^{\ell+1/2}(x). \quad (81)$$

Thus

$$u(\rho) = \rho^{\ell+1} e^{-\rho^2/2} L_k^{\ell+1/2}(\rho^2). \quad (82)$$

Returning to R :

$$R(r) = N_k r^\ell e^{-r^2/(2a^2)} L_k^{\ell+1/2}\left(\frac{r^2}{a^2}\right). \quad (83)$$

Step 9: Energy quantization

Recall from the previous step that the function $v(x)$ satisfies the associated Laguerre equation:

$$xv'' + \left(\ell + \frac{3}{2} - x\right)v' + kv = 0. \quad (84)$$

This differential equation has solutions in terms of associated Laguerre polynomials:

$$v(x) = L_k^{\ell+\frac{1}{2}}(x), \quad (85)$$

only if k is a non-negative integer:

$$k = 0, 1, 2, 3, \dots \quad (86)$$

Why must k be an integer? If k is not an integer, the solution becomes an infinite series that behaves like:

$$v(x) \sim e^x \quad (87)$$

for large x . Since our full solution contains

$$u(\rho) = \rho^{\ell+1} e^{-\rho^2/2} v(\rho), \quad (88)$$

this would give:

$$u(\rho) \sim e^{+\rho^2/2}, \quad (89)$$

which diverges as $\rho \rightarrow \infty$.

This is not a physically acceptable solution because the wavefunction is required to be normalizable:

$$\int |\Psi|^2 dV < \infty. \quad (90)$$

Therefore, we require the series to terminate, which happens when:

$$k \in \mathbb{Z}_{\geq 0} \quad (91)$$

We identify k as the **radial quantum number** and it determines the wavefunction's radial structure.

Solving for ϵ from Equation (78) gives:

$$\epsilon = k + \frac{2\ell + 3}{2}. \quad (92)$$

Recall that:

$$\epsilon = \frac{E}{\hbar\omega}. \quad (93)$$

Therefore, the allowed energy levels are

$$E = \hbar\omega \left(k + \frac{2\ell + 3}{2} \right) \quad (94)$$

$$= \hbar\omega \left(2k + \ell + \frac{3}{2} \right). \quad (95)$$

Interpretation:

- k is the radial quantum number.
- ℓ is the angular momentum quantum number.
- Energy depends on both k and ℓ .

We therefore define a **total quantum number**

$$\boxed{n = 2k + \ell} \quad (96)$$

so that

$$\boxed{E = \hbar\omega \left(n + \frac{3}{2} \right)} \quad (97)$$

This shows the energy quantization and degeneracy of the 3D quantum harmonic oscillator: different (k, ℓ) pairs can give the same energy.

A constraint upon ℓ naturally arises from Equation (96). This comes from

$$k = \frac{n - \ell}{2} \quad (98)$$

and the condition that k must be a non-negative integer, as seen in Equation (91). It follows that $n - \ell$ must be even. Therefore, n and ℓ must have the same parity. We now have a parity constraint upon ℓ :

$$\boxed{\ell = n, n - 2, n - 4, \dots \geq 0} \quad (99)$$

Step 10: Normalization

We now determine the normalization constant N_k such that:

$$\int_0^\infty r^2 |R(r)|^2 dr = 1. \quad (100)$$

Step 10.1: Substitute the radial solution

$$R(r) = N_k r^\ell e^{-r^2/(2a^2)} L_k^{\ell+\frac{1}{2}} \left(\frac{r^2}{a^2} \right). \quad (101)$$

Then:

$$|R|^2 = N_k^2 r^{2\ell} e^{-r^2/a^2} \left[L_k^{\ell+\frac{1}{2}} \left(\frac{r^2}{a^2} \right) \right]^2. \quad (102)$$

Multiply by r^2 :

$$r^2 |R|^2 = N_k^2 r^{2\ell+2} e^{-r^2/a^2} \left[L_k^{\ell+\frac{1}{2}} \left(\frac{r^2}{a^2} \right) \right]^2. \quad (103)$$

Step 9.2: Change variables

Let:

$$x = \frac{r^2}{a^2}. \quad (104)$$

Then:

$$r = a\sqrt{x}, \quad (105)$$

$$dr = \frac{a}{2\sqrt{x}} dx. \quad (106)$$

$$r^{2\ell+2} = (a\sqrt{x})^{2\ell+2} = a^{2\ell+2} x^{\ell+1}. \quad (107)$$

Step 9.3: Substitute everything

The normalization integral becomes:

$$\int_0^\infty r^2 |R|^2 dr = N_k^2 \int_0^\infty a^{2\ell+2} x^{\ell+1} e^{-x} \left[L_k^{\ell+\frac{1}{2}}(x) \right]^2 \cdot \frac{a}{2\sqrt{x}} dx. \quad (108)$$

Simplify powers of x :

$$x^{\ell+1} \cdot \frac{1}{\sqrt{x}} = x^{\ell+\frac{1}{2}}. \quad (109)$$

$$= N_k^2 \frac{a^{2\ell+3}}{2} \int_0^\infty x^{\ell+\frac{1}{2}} e^{-x} \left[L_k^{\ell+\frac{1}{2}}(x) \right]^2 dx. \quad (110)$$

Step 10.4: Use Laguerre identity

We use:

$$\int_0^\infty x^\alpha e^{-x} [L_k^\alpha(x)]^2 dx = \frac{\Gamma(k+\alpha+1)}{k!}. \quad (111)$$

Here:

$$\alpha = \ell + \frac{1}{2}. \quad (112)$$

Thus:

$$\int = \frac{\Gamma(k+\ell+\frac{3}{2})}{k!}. \quad (113)$$

Step 10.5: Solve for normalization constant

$$N_k^2 \frac{a^{2\ell+3}}{2} \cdot \frac{\Gamma(k+\ell+\frac{3}{2})}{k!} = 1. \quad (114)$$

Solve:

$$N_k^2 = \frac{2k!}{a^{2\ell+3} \Gamma(k+\ell+\frac{3}{2})}, \quad (115)$$

$$N_k = \sqrt{\frac{2k!}{a^{2\ell+3} \Gamma(k+\ell+\frac{3}{2})}}. \quad (116)$$

Remark: Gamma Function

The function $\Gamma(x)$ appearing in the normalization constant is known as the **Gamma function**. It is a generalization of the factorial to non-integer values and is defined by the integral

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt, \quad x > 0. \quad (117)$$

For positive integers, it satisfies

$$\Gamma(n+1) = n!, \quad (118)$$

so it naturally extends the factorial function.

In this problem, the normalization integral involves powers of $x^{\ell+1/2}$, which are generally not integers. As a result, factorials are not sufficient, and the Gamma function appears instead through the identity

$$\int_0^\infty x^\alpha e^{-x} [L_k^\alpha(x)]^2 dx = \frac{\Gamma(k+\alpha+1)}{k!}. \quad (119)$$

Thus, the Gamma function arises naturally when normalizing wavefunctions involving non-integer powers and ensures the result remains well-defined for all allowed quantum numbers.

Final normalized radial wavefunction:

$$R(r) = \sqrt{\frac{2k!}{a^{2\ell+3}\Gamma(k+\ell+\frac{3}{2})}} r^\ell e^{-r^2/(2a^2)} L_k^{\ell+\frac{1}{2}}\left(\frac{r^2}{a^2}\right) \quad (120)$$

Summary of Results

Full Solution

$$\Psi(r, \theta, \phi) = R(r)\Theta(\theta)\Phi(\phi) \quad (121)$$

Normalized Azimuthal Solution

$$\Phi(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi} \quad (122)$$

Polar Solution

$$\Theta(\theta) \propto P_\ell^m(\cos \theta) \quad (123)$$

Normalized Radial Solution

$$R(r) = \sqrt{\frac{2k!}{a^{2\ell+3}\Gamma(k+\ell+\frac{3}{2})}} r^\ell e^{-r^2/(2a^2)} L_k^{\ell+\frac{1}{2}}\left(\frac{r^2}{a^2}\right) \quad (124)$$

where $a = \sqrt{\frac{\hbar}{m\omega}}$.

Magnetic Quantum Number

$$m \in \mathbb{Z}, \quad |m| \leq \ell \quad (125)$$

Angular Momentum Quantum Number

$$\ell = n, n - 2, n - 4, \dots \geq 0 \quad (126)$$

Radial Quantum Number

$$k = \frac{n - \ell}{2} \quad (127)$$

Total Quantum Number

$$n = 2k + \ell \quad (128)$$

Energy Quantization

$$E = \hbar\omega \left(n + \frac{3}{2} \right) \quad (129)$$